

Cancer inhibition by inositol hexaphosphate (IP6) and inositol: from laboratory to clinic.

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Inositol hexaphosphate (IP6) is a naturally occurring polyphosphorylated carbohydrate that is present in substantial amounts in almost all plant and mammalian cells. It was recently recognized to possess multiple biological functions. A striking anticancer effect of IP6 was demonstrated in different experimental models. Inositol is also a natural constituent possessing moderate anticancer activity. The most consistent and best anticancer results were obtained from the combination of IP6 plus inositol. In addition to reducing cell proliferation, IP6 increases differentiation of malignant cells, often resulting in a reversion to normal phenotype. Exogenously administered IP6 is rapidly taken into the cells and dephosphorylated to lower-phosphate inositol phosphates, which further interfere with signal transduction pathways and cell cycle arrest. Enhanced immunity and antioxidant properties can also contribute to tumor cell destruction. However, the molecular mechanisms underlying this anticancer action are not fully understood. Because it is abundantly present in regular diet, efficiently absorbed from the gastrointestinal tract, and safe, IP6 holds great promise in our strategies for the prevention and treatment of cancer. IP6 plus inositol enhances the anticancer effect of conventional chemotherapy, controls cancer metastases, and improves the quality of life, as shown in a pilot clinical trial. The data strongly argue for the use of IP6 plus inositol in our strategies for cancer prevention and treatment. However, the effectiveness and safety of IP6 plus inositol at therapeutic doses needs to be determined in phase I and phase II clinical trials in humans.